

A contraction- and subdivision-closed class of graphs without additive improvement of quasi-isometries: a counterexample to Conjecture 1.2 of Nguyen, Scott and Seymour

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Abstract

We construct a fixed class of finite connected graphs, closed under edge contraction and edge subdivision, and a sequence of onto $(2, 1)$ -quasi-isometries to members of that class for which every additive $(1, A)$ approximation has $A \rightarrow \infty$. This disproves Conjecture 1.2 of Nguyen, Scott and Seymour as stated. The proof combines a rigid system of long complete-graph corridors with a red–blue turn-cost obstruction on high-girth 4-regular graphs.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 8](#) records the full provenance.

1 Introduction

All graphs in this note are finite, connected and simple. When an edge is contracted, loops and parallel edges are suppressed. The geometric realization of a graph gives every edge length one.

We use the convention of Nguyen, Scott and Seymour [1]: an (L, C) -quasi-isometry $f: G \rightarrow H$ satisfies

$$L^{-1}d_G(x, x') - C \leq d_H(f(x), f(x')) \leq Ld_G(x, x') + C$$

for all vertices x, x' , and every vertex of H is at distance at most C from $f(V(G))$. The last, coarse-surjectivity clause is essential here.

Conjecture 1.1 ([1, Conjecture 1.2]). Let $\mathcal{C}$ be a class of connected graphs, closed under contracting edges and subdividing edges. For all L, C there exists C' such that if there is an (L, C) -quasi-isometry from a graph G to a member of $\mathcal{C}$, then there is a $(1, C')$ -quasi-isometry from G to a member of $\mathcal{C}$.

Nguyen, Scott and Seymour prove the assertion when $\mathcal{C}$ is the class of graphs of bounded path-width [1, Theorem 1.3]. The same statement had appeared earlier as Conjecture 8 of Davies, Hatzel and Hickingbotham [2]. Their Theorem 4 disproves the stronger fixed-target version by a turn-cost construction. The obstruction below uses a related red–blue mechanism, but the long-corridor rigidity lemma is what permits both the map and the target to change.

Theorem 1.2 (Main theorem). *There is a class $\mathcal{C}$ of finite connected graphs, closed under edge contraction and edge subdivision, with the following property. For every integer $g \geq 10000$ there are finite connected graphs G_g and $H_g^+ \in \mathcal{C}$ and an onto $(2, 1)$ -quasi-isometry*

$$p_g: G_g \longrightarrow H_g^+$$

such that every $(1, A)$ -quasi-isometry from G_g to any member of $\mathcal{C}$ satisfies

$$A + 1 > \frac{g}{10000}.$$

Idea of proof. The class $\mathcal{C}$ consists of graphs in which every two branch vertices are joined by an internally degree-two path. The target H_g^+ is a subdivision of a complete graph: edges of a high-girth red–blue graph have length g , and all other edges have length $100g^2$. Splitting every original vertex into red and blue states produces G_g and the evident $(2, 1)$ map.

If an additive quasi-isometry from G_g to another member of $\mathcal{C}$ had small error, long-corridor rigidity would identify exactly one target branch vertex for each original vertex and exactly one target thread for each pair. The target would therefore induce edge lengths on the same complete-graph skeleton. Monochromatic paths force these lengths to be cheap on average, while an alternating path must pay one switch per turn in G_g . For a long enough path the two estimates contradict each other. [\[full proof\]](#)

Corollary 1.3. *Conjecture 1.1 is false, already for finite graphs and for $(L, C) = (2, 1)$.*

Proof. If [Conjecture 1.1](#) supplied a constant C' for the class in [Theorem 1.2](#), choose $g > 10000(C' + 1)$. The map p_g satisfies its premise, whereas its conclusion contradicts the lower bound on A in [Theorem 1.2](#). $\square$

2 The class $\mathcal{C}$

For a graph J , put

$$B(J) = \{v \in V(J) : \deg_J(v) \geq 3\}.$$

A *thread* between distinct $u, v \in B(J)$ is a u – v path all of whose internal vertices have degree two in J . Define

$$\mathcal{C} = \left\{ J : \begin{array}{l} J \text{ is finite, connected and simple, and every two distinct} \\ \text{vertices of } B(J) \text{ are joined by a thread} \end{array} \right\}. \quad (2.1)$$

Every subdivision of a complete graph belongs to $\mathcal{C}$.

Lemma 2.1. *The class $\mathcal{C}$ is closed under edge subdivision and edge contraction.*

Idea of proof. Subdivision preserves branch vertices and merely lengthens a thread. Under contraction no old degree can increase, and merging two vertices of degree at most two again gives degree at most two. The image of a thread between preimages of two new branch vertices contains the required thread; the only delicate case is when the contracted edge meets the interior of the old thread. [\[full proof\]](#)

Referee repair (contraction). The full proof separates the cases in which neither, one, or both ends of the contracted edge lie on the old thread. In particular, an edge from an internal thread vertex to a vertex outside the thread would give that internal vertex degree at least three, so the missing chord case cannot occur.

3 Metric preliminaries

Lemma 3.1 (Continuous coarse inverses). *If there is a $(1, A)$ -quasi-isometry between graphs G and K , then their geometric realizations X, Y admit continuous maps*

$$F: X \rightarrow Y, \quad Q: Y \rightarrow X$$

such that, with $D = 10(A + 1)$,

$$\begin{aligned} |d_Y(Fx, Fx') - d_X(x, x')| &\leq D, \\ |d_X(Qy, Qy') - d_Y(y, y')| &\leq D, \\ d_Y(FQy, y) &\leq D. \end{aligned} \quad (3.1)$$

Idea of proof. Extend the vertex map and a coarse inverse along geodesics. Comparing arbitrary points with endpoints of their unit edges yields distortion bounds $3A + 4$ and $9A + 4$ and a closeness bound $7A + 6$, all at most D . [\[full proof\]](#)

Referee repair (constants). The appendix derives $3A + 4$, $9A + 4$, and $7A + 6$ explicitly rather than asserting them.

Lemma 3.2 (Detecting a branch vertex). *Let Z be a metric graph. Suppose z_1, z_2, z_3 lie within distance r of z and*

$$d_Z(z_i, z_j) > r \quad (i \neq j).$$

Then the closed radius- r ball about z meets $B(Z)$.

Idea of proof. If the ball has maximum degree at most two, the union of three shortest paths from z lies in a path or in a cycle component. A path puts two points on one side of z ; a cycle component created inside the ball has length at most $2r$. Either way two of the three points are at distance at most r . [\[full proof\]](#)

Referee repair (cycle case). If the union closes into a cycle while every vertex in the ball has degree at most two, that cycle is an entire connected component of Z . This justifies the diameter bound used in the argument.

4 Rigidity of long complete-graph corridors

Lemma 4.1 (Long-corridor rigidity). *Let $n \geq 4$, $D \geq 1$, and $S \geq 1000D$. Construct a metric graph X by replacing each vertex v of K_n by a tree T_v of diameter at most one, choosing $r_v \in T_v$, and joining T_u to T_v by one corridor I_{uv} of length $L_{uv} \geq S$ for each pair u, v . Assume that corridor interiors are disjoint and there are no other edges.*

Let Y be the geometric realization of a graph in $\mathcal{C}$, and suppose $F: X \rightarrow Y$ and $Q: Y \rightarrow X$ satisfy (3.1). Then:

- (i) *Y has exactly n branch vertices, labelled h_v ;*
- (ii) *exactly one thread joins h_u to h_v for each pair u, v ;*
- (iii) *if λ_{uv} is its length and d_λ is the weighted complete-graph metric, then*

$$|d_\lambda(u, v) - d_X(r_u, r_v)| \leq 30D; \tag{4.1}$$

- (iv) *every thread satisfies*

$$\lambda_{uv} \geq \frac{L_{uv}}{2} - D. \tag{4.2}$$

Idea of proof. Three well-separated points near each r_v detect a target branch vertex h_v . Conversely, three long target threads from any branch vertex detect a unique source cluster T_v . After trimming $50D$ from both ends, a target thread cannot approach any source cluster and its image under Q must remain in one corridor; continuity forces it to cover that corridor's central part. Two threads for the same cluster pair would then give points far apart in Y but D -close under Q . Hence clusters and threads are unique, and the distortion estimates give (4.1)–(4.2). [\[full proof\]](#)

Referee repair (induced metric). After uniqueness is established, every component outside the complete system of threads meets it in at most one branch vertex; such a pendant path or cycle cannot shorten a path between branch vertices. Thus their metric is exactly the weighted K_n metric d_λ .

5 High-girth red–blue graphs

Lemma 5.1. *For every integer $g \geq 5$ there is a finite connected simple 4-regular graph H_g of girth at least g whose edge set is the disjoint union $E_R \sqcup E_B$, with two incident red edges and two incident blue edges at every vertex. Every monochromatic component is a cycle of length at least g .*

Idea of proof. Use a finite quotient of the free group on a, b in which no nonempty reduced word of length less than g vanishes. Its Cayley graph for $a^{\pm 1}, b^{\pm 1}$ has the required girth, and the two generators give the colors. A finite quotient is obtained explicitly by completing the partial permutations traced by every short reduced word. [\[full proof\]](#)

6 Construction and quantitative obstruction

Fix $g \geq 10000$, let $H = H_g$ be supplied by [Lemma 5.1](#), put

$$n = |V(H)|, \quad S = g, \quad M = 100g^2. \quad (6.1)$$

Construct H_g^+ from K_n on vertex set $V(H)$ by replacing every edge of H by a path of length S , and every nonedge of H by a path of length M . Thus $H_g^+ \in \mathcal{C}$.

Construct G_g by splitting each original vertex v into v_R, v_B , joined by a unit switch edge. Attach red H -corridors to the R vertices, blue H -corridors to the B vertices, and all length- M corridors to the R vertices.

Lemma 6.1. *Collapsing every switch edge and mapping each corridor identically defines an onto $(2, 1)$ -quasi-isometry*

$$p_g: G_g \longrightarrow H_g^+.$$

Idea of proof. Projection never increases length. A shortest target path of length k lifts by traversing the same corridor edges and using at most $k + 1$ switch edges, so the lifted distance is at most $2k + 1$. [\[full proof\]](#)

Lemma 6.2 (Turn-cost obstruction). *Every $(1, A)$ -quasi-isometry from G_g to a member of $\mathcal{C}$ satisfies*

$$A + 1 > \frac{g}{10000}.$$

Idea of proof. Assume the reverse inequality and put $D = 10(A + 1)$, $E = 30D$, and $s = \lfloor g/10 \rfloor$. By [Lemma 4.1](#), the target induces lengths λ_e on K_n approximating distances between the R states. Every monochromatic s -path has λ -length at most $Ss + 2 + E$. Averaging cyclic windows over all monochromatic cycles therefore produces an alternating s -path with the same upper bound. Its unique lift in G_g needs $s - 1$ switches and has length at least $Ss + s - 1$, giving $s - 3 \leq 2E$, contrary to the chosen constants. [\[full proof\]](#)

7 Remarks

Remark 7.1 (Coarse surjectivity). The coarse-surjectivity clause in the definition is load-bearing. Without it one could attach to G_g a thread between every pair of its branch vertices that does not already have one; the resulting graph lies in $\mathcal{C}$ and contains G_g isometrically. Thus the result does not concern quasi-isometric embeddings. The definition in [\[1\]](#) does include coarse surjectivity.

Remark 7.2 (Scope). The class $\mathcal{C}$ is intentionally economical: it is not closed under edge deletion, disjoint union, or taking minors, and it does not contain every tree. The result refutes the universally quantified conjecture with exactly its stated closure assumptions. It leaves open variants for deletion-closed or minor-closed classes and the planar case highlighted in [\[2\]](#).

Remark 7.3 (Computational audit). The referee checked closure on all 995 connected graphs with at most seven vertices; 381 belonged to $\mathcal{C}$, and every edge contraction and subdivision stayed in $\mathcal{C}$. A further 20 000 random larger members gave no failure. The quasi-isometry inequalities were checked exhaustively on three finite stand-ins with $(S, M) = (3, 5), (4, 9), (6, 11)$, having respectively 1248, 2392, and 3042 vertices in G_g . The branch-detection lemma passed 16 300 tested instances. The turn-cost linear program attained exactly $(s - 3)/2$ for $s = 2, \dots, 6$ whenever the finite test graph contained the required paths. These computations do not test [Lemma 4.1](#) in its general continuous form, which the referee checked only line by line.

Remark 7.4 (Constants). The threshold and the factor 10000 are deliberately loose. The referee verified that the numerical block of inequalities in the proof already holds from $g = 80$. No optimization is claimed.

8 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The counterexample was found by gpt-6-astra, reasoning effort max, in a single pass begun on 2026-09-17, catalog id 2509.09031__00, leg attacks_retry. This was a second attempt; the earlier gpt-5.6-sol writeup and its partial verdict were supplied as context. The earlier writeup gave an infinite-graph entropy counterexample and a finite fixed-map obstruction, neither of which settled the finite arbitrary-map statement. Its red–blue turn-cost idea was reused, while the long-corridor rigidity argument is new to the retry.

(2) An adversarial referee agent (claude-opus-5) audited the writeup on 2026-09-17 and returned MINOR GAPS with medium confidence. It checked the source definition and quantifiers, re-derived every inequality, compared the turn-cost mechanism with Davies–Hatzel–Hickingbotham [2], searched for priority, and ran the closure, construction, metric, high-girth, and linear-program computations summarized above.

(3) A rewrite was started by claude-fable-5-1 on 2026-09-18 but stopped at a compilable skeleton when that session hit its rate limit. The interrupted transcript and skeleton were recovered, and Codex completed the present note on 2026-09-18, with short proof ideas in the main text and full proofs in [Appendix A](#). Relative to the original writeup it makes the following declared changes: it restricts $\mathcal{C}$ explicitly to connected graphs; cites the prior fixed-target obstruction [2]; explains why coarse surjectivity is essential; checks $n \geq 4$ and $D \geq 1$ at the point of use; expands the contraction case analysis in [Lemma 2.1](#); derives all constants in [Lemma 3.1](#); repairs the cycle-component sentence in [Lemma 3.2](#); and explains why components outside the unique thread system cannot shorten its metric. No other mathematical content was changed. **No human mathematician has checked this argument.**

A Full proofs

Lemma 2.1. *The class $\mathcal{C}$ is closed under edge subdivision and edge contraction.*

Proof. Subdividing an edge leaves all old degrees unchanged and gives the new vertex degree two. Hence the branch set is unchanged, and replacing an edge of any old thread by the corresponding two-edge path leaves a thread.

Now let J' be obtained from $J \in \mathcal{C}$ by contracting xy to z . Degrees of vertices outside $\{x, y\}$ do not increase after loops and parallel edges are suppressed. If both x and y have degree at most two, then

$$\deg_{J'}(z) \leq (\deg_J(x) - 1) + (\deg_J(y) - 1) \leq 2.$$

Thus every branch vertex of J' is the image of at least one branch vertex of J .

Take distinct $u', v' \in B(J')$ and branch vertices $u, v \in B(J)$ mapping to them. Let P be a thread from u to v . If neither x nor y lies on P , its image is unchanged. If exactly one, say x , lies internally on P , then the edge xy would be a third edge at x unless y is one of the two neighbours of x on P ; hence y also lies on P , a contradiction. Thus an edge with only one endpoint on P can meet P only at an endpoint, and contraction merely shortens its first or last edge.

If both x, y lie on P , then either they are consecutive on P , in which case their merger shortens P , or xy is a chord. A chord between nonconsecutive vertices cannot be incident with an internal vertex of P , because that vertex already has its two path neighbours and would have degree at least three. Since u' and v' are distinct, the only remaining endpoint cases again leave a $u'-v'$ walk containing a simple path. Every internal vertex of that simple path is the image of a degree-two internal vertex or the merger of two degree-two vertices and therefore has degree at most two. It has degree at least two because it is internal on the simple path, hence degree exactly two. The path is a thread in J' . $\square$

Lemma 3.1 (Continuous coarse inverses). *If there is a $(1, A)$ -quasi-isometry between graphs G and K , then their geometric realizations X, Y admit continuous maps*

$$F: X \rightarrow Y, \quad Q: Y \rightarrow X$$

such that, with $D = 10(A + 1)$,

$$\begin{aligned} |d_Y(Fx, Fx') - d_X(x, x')| &\leq D, \\ |d_X(Qy, Qy') - d_Y(y, y')| &\leq D, \\ d_Y(FQy, y) &\leq D. \end{aligned} \tag{3.1}$$

Proof. Let $f: V(G) \rightarrow V(K)$ be a $(1, A)$ -quasi-isometry. Extend f continuously over each unit edge of G along a shortest path in K ; call the extension F . Since adjacent vertices have images at distance at most $A + 1$, each point x of an edge with endpoint v satisfies $d_Y(Fx, f(v)) \leq A + 1$. Given x, x' and chosen incident endpoints v, v' ,

$$\begin{aligned} d_Y(Fx, Fx') &\leq (A + 1) + d_K(fv, fv') + (A + 1) \\ &\leq d_G(v, v') + 3A + 2 \\ &\leq d_X(x, x') + 3A + 4. \end{aligned}$$

The same endpoint comparison and the lower quasi-isometry inequality give the reverse estimate with the same error.

For every vertex w of K , coarse surjectivity supplies $q(w) \in V(G)$ with $d_K(fq(w), w) \leq A$. For vertices w, w' ,

$$\begin{aligned} d_G(qw, qw') &\leq d_K(fq(w), fq(w')) + A \leq d_K(w, w') + 3A, \\ d_G(qw, qw') &\geq d_K(fq(w), fq(w')) - A \geq d_K(w, w') - 3A. \end{aligned}$$

Extend q over each edge of K along a geodesic. The image of such an edge has length at most $3A + 1$. Comparing arbitrary points to endpoints adds at most $2(3A + 1) + 2$, giving distortion at most $9A + 4$.

Finally choose an endpoint w of the edge containing y . Then

$$\begin{aligned} d_Y(FQy, y) &\leq d_Y(FQy, Fq(w)) + d_Y(fq(w), w) + d_Y(w, y) \\ &\leq (d_X(Qy, q(w)) + 3A + 4) + A + 1 \\ &\leq (3A + 1) + 3A + 4 + A + 1 = 7A + 6. \end{aligned}$$

For $A \geq 0$, all three bounds are at most $10(A + 1) = D$. $\square$

Lemma 3.2 (Detecting a branch vertex). *Let Z be a metric graph. Suppose z_1, z_2, z_3 lie within distance r of z and*

$$d_Z(z_i, z_j) > r \quad (i \neq j).$$

Then the closed radius- r ball about z meets $B(Z)$.

Proof. Suppose the closed r -ball about z contains no branch vertex. Choose shortest paths P_i from z to z_i . Their union has maximum degree at most two. If it is contained in a path, deleting z leaves at most two directions, so two z_i lie in the same direction and their distance is at most the larger of their distances from z , hence at most r .

Otherwise the union contains a cycle. Since every vertex of the cycle lies in the ball and has degree at most two in the whole graph, no edge leaves the cycle; it is an entire connected component of Z . The cycle is formed from two paths of length at most r , so its length is at most $2r$ and its diameter at most r . Again two of the z_i are at distance at most r , a contradiction. $\square$

Lemma 4.1 (Long-corridor rigidity). *Let $n \geq 4$, $D \geq 1$, and $S \geq 1000D$. Construct a metric graph X by replacing each vertex v of K_n by a tree T_v of diameter at most one, choosing $r_v \in T_v$, and joining T_u to T_v by one corridor I_{uv} of length $L_{uv} \geq S$ for each pair u, v . Assume that corridor interiors are disjoint and there are no other edges.*

Let Y be the geometric realization of a graph in $\mathcal{C}$, and suppose $F: X \rightarrow Y$ and $Q: Y \rightarrow X$ satisfy (3.1). Then:

- (i) *Y has exactly n branch vertices, labelled h_v ;*
- (ii) *exactly one thread joins h_u to h_v for each pair u, v ;*
- (iii) *if λ_{uv} is its length and d_λ is the weighted complete-graph metric, then*

$$|d_\lambda(u, v) - d_X(r_u, r_v)| \leq 30D; \quad (4.1)$$

- (iv) *every thread satisfies*

$$\lambda_{uv} \geq \frac{L_{uv}}{2} - D. \quad (4.2)$$

Proof. Step 1: branch vertices near the source clusters. For each v , choose three distinct corridors incident with T_v . On each choose a point at corridor-distance $10D$ from T_v . Each is within $10D + 1 \leq 12D$ of r_v , while every pair is at distance at least $20D$. Their F -images lie within $12D$ of $F(r_v)$ and are pairwise at distance at least $19D$. Lemma 3.2 with radius $12D$ gives

$$h_v \in B(Y), \quad d_Y(h_v, F(r_v)) \leq 12D. \quad (A.1)$$

For $u \neq v$,

$$d_Y(h_u, h_v) \geq S - D - 24D = S - 25D. \quad (A.2)$$

Step 2: every target branch vertex has a unique cluster. Let $h \in B(Y)$. By (A.2), at most one h_v lies within $S/3$ of h ; since $n \geq 4$, choose three others. The defining property of $\mathcal{C}$ supplies threads from h to them. Their interiors are disjoint, so points at thread-distance $10D$ from h are pairwise $20D$ apart. Their Q -images lie within $11D$ of $Q(h)$ and are pairwise at least $19D$ apart. Lemma 3.2 in X gives a branch vertex within $11D$ of $Q(h)$. Every branch vertex of X belongs to some T_v , so

$$d_X(Qh, r_v) \leq 12D. \quad (A.3)$$

Using (3.1) and (A.1),

$$d_Y(h, F(r_v)) \leq 14D. \quad (A.4)$$

The cluster v is unique because distinct $F(r_v)$ are at distance at least $S - D > 28D$.

Step 3: a thread follows the corresponding corridor. Let P be a thread of length λ between branch vertices in clusters v, w . It has $\lambda \geq S - 29D > 100D$. Trim the first and last $50D$, leaving P^0 . Every $y \in P^0$ is at distance at least $50D$ from $B(Y)$. If $d_X(Qy, r_z) \leq 20D$, then

$$d_Y(y, h_z) \leq D + 21D + 12D = 34D,$$

a contradiction. Hence $Q(P^0)$ avoids the closed $20D$ -balls about the r_z and, by continuity, lies in one trimmed corridor component.

If y_0, y_1 are the ends of P^0 , then

$$d_X(Qy_0, r_v), d_X(Qy_1, r_w) \leq 50D + D + 12D = 63D.$$

As $63D < S$, the component is I_{vw} . Continuity shows that $Q(P^0)$ covers

$$[63D, L_{vw} - 63D]. \tag{A.5}$$

Step 4: uniqueness. Two distinct threads joining the same clusters would, by (A.5), contain points mapping to the midpoint of I_{vw} . Their Y -distance would be at most D , although leaving either trimmed thread costs at least $50D$. Thus there is at most one such thread. If a cluster contained two branch vertices, their threads to another cluster would contradict this. Hence each cluster contains exactly one branch vertex and $\mathcal{C}$ supplies exactly one thread for each pair.

Step 5: metric estimates. Every maximal degree-two chain joining distinct branch vertices is one of these threads. Any remaining component meets the thread system in at most one branch vertex, so entering it cannot shorten a branch-to-branch path. Their metric is therefore d_λ . From (A.1) and the distortion of F ,

$$|d_\lambda(u, v) - d_X(r_u, r_v)| \leq 25D,$$

which implies (4.1).

Finally (A.5) contains the quarter and three-quarter points of I_{uv} . Their X -distance is $L_{uv}/2$. Choose preimages on the thread; the distortion of Q gives their Y -distance at least $L_{uv}/2 - D$, which is at most λ_{uv} . This proves (4.2). $\square$

Lemma 5.1. *For every integer $g \geq 5$ there is a finite connected simple 4-regular graph H_g of girth at least g whose edge set is the disjoint union $E_R \sqcup E_B$, with two incident red edges and two incident blue edges at every vertex. Every monochromatic component is a cycle of length at least g .*

Proof. Let W_g be the set of all nonempty freely reduced words of length less than g in a, a^{-1}, b, b^{-1} . For a word w of length k , use $\{0, \dots, k\}$ and prescribe partial permutations a_w, b_w so that reading w follows $0 \rightarrow 1 \rightarrow \dots \rightarrow k$; an inverse letter prescribes the reverse arrow. These partial maps are injective and well-defined, since a conflict at an intermediate point would require adjacent inverse letters. Complete them to permutations.

In the product of the resulting finite symmetric groups, let a, b denote the tuples and let $Q_g = \langle a, b \rangle$. No nonempty reduced word of length less than g is trivial in Q_g , because its own coordinate sends 0 to its length. The undirected Cayley graph of Q_g for $a^{\pm 1}, b^{\pm 1}$ is finite, connected, simple, 4-regular and has girth at least g . Color a -edges red and b -edges blue. Each color has degree two at every vertex, hence is a union of cycles; a monochromatic cycle shorter than g would give a forbidden reduced relation. $\square$

Lemma 6.1. *Collapsing every switch edge and mapping each corridor identically defines an onto $(2, 1)$ -quasi-isometry*

$$p_g: G_g \longrightarrow H_g^+.$$

Proof. The map is onto. Projection collapses switch edges and is isometric on corridors, so

$$d_{H_g^+}(p_g x, p_g y) \leq d_{G_g}(x, y).$$

Conversely, lift a shortest target path edge by edge. Each corridor edge costs one in G_g ; at most one switch is needed between consecutive corridor edges and at either endpoint. A target path of length k therefore lifts with at most $k + 1$ switches:

$$d_{G_g}(x, y) \leq 2d_{H_g^+}(p_g x, p_g y) + 1.$$

□

Lemma 6.2 (Turn-cost obstruction). *Every $(1, A)$ -quasi-isometry from G_g to a member of $\mathcal{C}$ satisfies*

$$A + 1 > \frac{g}{10000}.$$

Proof. Suppose $A + 1 \leq g/10000$. Set

$$D = 10(A + 1), \quad E = 30D, \quad s = \left\lfloor \frac{g}{10} \right\rfloor. \quad (\text{A.6})$$

Then $D \geq 1$, $S = g \geq 1000D$, and $n \geq 4$. Apply [Lemmas 3.1](#) and [4.1](#) to G_g , with switch edges as T_v and $r_v = v_R$. We obtain lengths λ_{uv} on K_n such that

$$|d_\lambda(u, v) - d_{G_g}(u_R, v_R)| \leq E \quad (\text{A.7})$$

and, for $uv \notin E(H)$,

$$\lambda_{uv} \geq M/2 - D. \quad (\text{A.8})$$

For $g \geq 10000$ and $D \leq g/1000$,

$$\begin{aligned} s - 3 &> 2E, & Ss &> 2 + 2E, & S &> s + 1, \\ M/2 - D &> Ss + 2 + E, & 2s &< g/2, & g - s &\geq 2s. \end{aligned} \quad (\text{A.9})$$

If P is an H -path of length $t < g/2$, it is geodesic. A G_g -path between the corresponding R states either uses a long corridor, costing at least $M > St$, or at least t short corridors. Hence

$$\lambda(P) \geq d_\lambda(u, v) \geq St - E. \quad (\text{A.10})$$

Let P be a monochromatic H -path of length s . A red path lifts between R states with length Ss , and a blue path with length $Ss + 2$. Thus $d_\lambda(u, v) \leq Ss + 2 + E$. A λ -shortest path Q cannot use a nonedge by (A.8)–(A.9). If $Q \neq P$, then $P \cup Q$ contains a cycle, so Q contains a $2s$ -edge subpath. Equation (A.10) gives $\lambda(Q) \geq 2Ss - E > Ss + 2 + E$, a contradiction. Therefore

$$\lambda(P) \leq Ss + 2 + E. \quad (\text{A.11})$$

Apply (A.11) to every cyclic window of s edges on every monochromatic cycle. Every edge occurs in exactly s windows, so

$$\sum_{e \in E(H)} \lambda_e \leq |E(H)| \left(S + \frac{2 + E}{s} \right). \quad (\text{A.12})$$

Choose a uniformly random directed edge and continue for s edges by alternating colors, choosing uniformly one of the two outward edges of the required color. At every position the directed edge remains uniform. Thus some alternating walk $P = v_0 \cdots v_s$ satisfies

$$\lambda(P) \leq Ss + 2 + E. \quad (\text{A.13})$$

It is nonbacktracking; since $s < g$ it is a path, and since $2s < g$ it is the unique H -geodesic between its endpoints.

A shortest G_g -path between $(v_0)_R$ and $(v_s)_R$ cannot use a long corridor because $M > Ss + s + 1$, nor $s + 1$ short corridors because $S(s + 1) > Ss + s + 1$. It uses exactly s short corridors, projects to P , and needs at least $s - 1$ switches:

$$d_{G_g}((v_0)_R, (v_s)_R) \geq Ss + s - 1. \quad (\text{A.14})$$

Equations (A.7), (A.13), and (A.14) yield $s - 3 \leq 2E$, contrary to (A.9). $\square$

Theorem 1.2 (Main theorem). *There is a class $\mathcal{C}$ of finite connected graphs, closed under edge contraction and edge subdivision, with the following property. For every integer $g \geq 10000$ there are finite connected graphs G_g and $H_g^+ \in \mathcal{C}$ and an onto $(2, 1)$ -quasi-isometry*

$$p_g: G_g \longrightarrow H_g^+$$

such that every $(1, A)$ -quasi-isometry from G_g to any member of $\mathcal{C}$ satisfies

$$A + 1 > \frac{g}{10000}.$$

Proof. Take $\mathcal{C}$ from (2.1). It is closed under the two operations by Lemma 2.1. For each $g \geq 10000$, the construction gives $H_g^+ \in \mathcal{C}$ and G_g . Lemma 6.1 supplies the onto $(2, 1)$ -quasi-isometry and Lemma 6.2 the asserted lower bound. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its section, lemma and equation numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/2509.09031__00/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The target is Conjecture 1.2 of Nguyen–Scott–Seymour, *Asymptotic structure. II* (arXiv:2509.09031). I fetched the paper and the statement in `prompt.md` is verbatim:

“Let $\mathcal{C}$ be a class of connected graphs, closed under contracting edges and subdividing edges. For all L, C there exists C' such that if there is an (L, C) -quasi-isometry from a graph G to a member of $\mathcal{C}$, then there is a $(1, C')$ -quasi-isometry from G to a member of $\mathcal{C}$.”

The same statement is Conjecture 8 of Davies–Hatzel–Hickingbotham (arXiv:2503.07448), where it is explicitly posed as the surviving weakening after they refuted the fixed-target version.

Three interpretation points, all of which come out in the writeup’s favour:

1. **Coarse surjectivity.** The writeup’s caveat block flags that it “uses the standard coarsely onto definition of quasi-isometry”. This is load-bearing and it is the right choice. NSS’s own definition (Section 1 of 2509.09031) has three clauses, the third being “for every $y \in V(H)$ there exists $v \in V(G)$ such that $\text{dist}_H(\varphi(v), y) \leq C$ ”. So the intended reading is the one used. This matters enormously: **without** coarse surjectivity the counterexample dies outright, because one can attach, to the graph G_g itself, a thread of length 10^9 between every pair of its branch vertices that does not already have one; the result is a member of $\mathcal{C}$ into which G_g embeds isometrically, i.e. by a $(1, 0)$ -quasi-isometric embedding. The writeup does not spell this out but it does flag the dependence.
2. **Quantifier form.** The writeup fixes $(L, C) = (2, 1)$ and produces, for each g , a graph G_g that is $(2, 1)$ -quasi-isometric to a member of $\mathcal{C}$ but admits no $(1, A)$ -quasi-isometry to any member of $\mathcal{C}$ with $A + 1 \leq g/10000$. That is exactly the negation of Conjecture 1.2. The QI convention difference (NSS use $d_G \leq L \cdot d_H + C$ where the writeup uses $L^{-1}d_G - C \leq d_H$) is immaterial: for $L = 1$ both say $|d_G - d_H| \leq C$, and my computation verified the positive direction in the *stronger* form $d_{H+} \leq d_G$ and $d_G \leq 2d_{H+} + 1$, which satisfies both conventions at $(2, 1)$.
3. **Is $\mathcal{C}$ a strawman class?** $\mathcal{C} = \{J : \text{every two distinct vertices of degree } \geq 3 \text{ are joined by a path whose internal vertices have degree } 2\}$. It is contrived but it is a legitimate instance: the conjecture is universally quantified over classes and imposes only the two closure properties. It is *not* closed under edge deletion, disjoint union, or minors, and the writeup says so. NSS’s motivating remarks give no indication that they intended extra hypotheses, and DHH single out the planar case only as “of particular interest”. I would call this a fair refutation of the conjecture as stated, though an author could reasonably respond by re-asking it for classes that are also deletion-closed (the writeup itself says this version is untouched).

B.2 Step-by-step findings

step	label	note
§1 def. of $\mathcal{C}$ (1.1)	VALID	Well-defined; contains every subdivision of a complete graph, hence is non-trivial.
Lemma 1.1, subdivision closure	VALID	$B(J)$ is unchanged and every thread survives (lengthened). Verified exhaustively: 381 members of $\mathcal{C}$ among all 995 connected graphs on ≤ 7 vertices, plus 20 000 random larger members — 0 failures.
Lemma 1.1, contraction closure	MINOR GAP (repaired)	The key facts — degrees of uncontracted vertices never increase; merging two vertices of degree ≤ 2 gives degree ≤ 2 ; hence $B(J') \subseteq \text{image of } B(J)$ — are correct. The sentence “After choosing a simple path in the image, its internal vertices have degree exactly two” hides one case: if the contracted edge xy is a chord of the thread P with x internal, one must argue x would then have degree ≥ 3 , contradiction; and if x,y are consecutive interior vertices the image really is a path. I worked the full case analysis (x,y both internal / one an endpoint / neither on P) and it closes. Brute force confirms: 0 failures over $\sim 20\,400$ members of $\mathcal{C}$ and every edge of each.
Lemma 2.1 (continuous coarse inverses)	MINOR GAP (repaired)	The three distortion constants ($3A+4$ for F , $9A+4$ for Q , $7A+6$ for FQ) are asserted, not derived. I re-derived all three and they hold; each is $\leq 10(A+1) = D$ for $A \geq 0$. VALID after the derivation.
Lemma 2.2 (detecting a branch vertex)	MINOR GAP (repaired)	Correct, but the cycle case needs one extra sentence: if every vertex within distance r of z has degree ≤ 2 , the cycle formed is a whole connected component of Z , which is what makes “length $\leq 2r$ hence diameter $\leq r$ ” legitimate. Brute-forced: 16 300 instances of the hypothesis over 4 000 random graphs, 0 violations of the conclusion.
Lemma 3.1 Step 1 (branch vertex near each $F(r_v)$)	VALID	Three corridor points at corridor-distance $10D$ from T_v are at pairwise X -distance $\geq 20D$ and within $10D+1$ of r_v (uses $\text{diam } T_v \leq 1$); F pushes these to $\geq 19D$ apart and $\leq 12D$ from $F(r_v)$, so Lemma 2.2 with $r = 12D$ applies ($19D > 12D$). (3.4): $d_Y(h_u, h_v) \geq S - 25D$. Checked.
Lemma 3.1 Step 2 (every branch vertex is in a cluster)	VALID	Needs $n \geq 4$ to pick three h_v at distance $\geq S/3$ (fine, since pairwise distances exceed $2S/3$). The three threads from h leave by distinct edges and hence have disjoint interiors — correct because an interior vertex of a thread has degree 2, so a thread is the <i>unique</i> maximal degree-2 chain in its direction. Points at thread-distance $10D$ are at Y -distance exactly $10D$ and pairwise $20D$ (uses thread length $\geq S/3 \gg 10D$). Lemma 2.2 in X with $r = 11D$ gives a branch vertex of X , and all of $B(X)$ lies in the trees T_v . Uniqueness of v from $S - D > 28D$. All arithmetic re-derived and correct.
Lemma 3.1 Step 3 (a thread follows its corridor)	VALID	The trimming by $50D$ at each end is legitimate since $\lambda \geq S - 29D > 100D$. (3.7) is correct: if $d_X(Qy, r_z) \leq 20D$ then $d_Y(y, h_z) \leq D + 21D + 12D = 34D < 50D$. The components of X minus the $20D$ -balls are exactly the trimmed corridor interiors (each corridor has length $\geq S \geq 1000D$). Endpoint bound $50D + D + 12D = 63D < S$ forces the corridor to be I_{vw} , and continuity gives the covering (3.8).

step	label	note
Lemma 3.1 Step 4 (uniqueness of threads/clusters)	VALID	Two threads to the same cluster pair would both cover the midpoint of I_{vw} under Q , forcing $d_Y(y, y') \leq D$ between points on threads with disjoint interiors each $\geq 50D$ from $B(Y)$ — impossible. The extension to “one branch vertex per cluster” is correct and uses $n \geq 2$.
Lemma 3.1 Step 5 (metric + (3.2))	MINOR GAP (repaired)	“Other components — pendant paths or cycles attached at a single branch vertex — cannot shorten such a path” is one line but correct: every maximal degree-2 chain between two <i>distinct</i> branch vertices is a thread and there is exactly one per pair, so the branch-vertex metric is exactly the weighted K_n metric. The derivation
§4 high-girth 2-coloured graph	VALID	The partial-permutation recipe is exactly the standard residual-finiteness separation. The stated well-definedness reason (“a conflict at an intermediate index would require adjacent inverse letters”) is right. Implemented and checked on every nonempty reduced word of length $< g$ for $g = 4, 5, 6, 7$ (1 456 words at $g = 7$): the word’s own coordinate sends $0 \mapsto H_g^+$ is a subdivision of K_n , hence in $\mathcal{C}$. The lift bound “at most $k+1$ switches” is crude but valid. Verified exhaustively on three instances (see below): p is onto and satisfies $d_{\{H^+\}} \leq d_G$ and $d_G \leq 2 d_{\{H^+\}} + 1$ with equality attained.
§5 G_g, H_g^+ , and p is an onto (2,1)-QI	VALID (computationally verified)	Lemma 3.1 applies with $T_v = \text{switch edge}$ (a tree of diameter 1), $r_v = v_R, S = g \geq 1000D \iff A+1 \leq g/10000$, and $n \geq 4$. The hypothesis $n \geq 4$ is never explicitly checked in the writeup (it holds trivially). All nine inequalities of block (6.5) verified numerically; they first all hold at $g = 80$, so the stated threshold $g \geq 10000$ is comfortable.
§6 setup (6.1)–(6.5)	VALID	$d_G(u_R, v_R) \geq St$ for H -paths of length $t < g/2$: a long corridor costs $M = 100g^2 > St$, and any route through H -corridors uses $\geq d_H(u, v) = t$ of them. Verified on the test instance.
§6.1 lower bound (6.6)	VALID	Red monochromatic s -paths lift with no switches (cost Ss), blue with two (cost $Ss+2$). The λ -shortest path cannot leave $E(H)$ because (6.4) forces $\lambda \geq M/2 - D$ on non-edges, and cannot differ from P because a cycle of length $\geq g$ would force a $2s$ -subpath of λ -length $\geq 2Ss - E > Ss+2+E$. Both checks re-derived.
§6.2 upper bound (6.8)	VALID	Cyclic-window counting verified (every edge of a cycle lies in exactly s of the L windows). The alternating non-backtracking walk really does preserve the uniform measure on directed edges — each directed edge has exactly two opposite-colour predecessors, each transitioning with probability $1/2$; verified numerically to machine zero over 6 steps on a concrete 2-factorised 4-regular graph.
§6.3 averaging (6.9)–(6.10)	VALID	A shortest G -path is a path, so it traverses each corridor it enters in full; it cannot use a long corridor ($M > Ss+s+1$) nor $s+1$ short ones ($S(s+1) > Ss+s+1$), so it uses exactly s and projects onto the unique H -geodesic P ; alternation forces $s-1$ switches. The final chain gives $s-3 \leq 2E$, contradicting (6.5). Verified as an LP (below): the optimum is exactly $(s-3)/2$.
§6.4 switch cost (6.11) and the contradiction	VALID	The negation of Conjecture 1.2 with $(L, C) = (2, 1)$ follows.
§7 conclusion	VALID	

I found **no ERROR**. The gaps above are all of the “one missing sentence” kind and I closed each of them.

B.3 Reference check

- **arXiv:2509.09031, Nguyen–Scott–Seymour, *Asymptotic structure. II*** — fetched. Conjecture 1.2 is verbatim as in prompt.md. The definition of (L,C)-quasi-isometry has three clauses including coarse surjectivity, which is what the writeup assumes. Theorem 1.3 is the bounded-path-width special case. Confirmed.
- **arXiv:2503.07448, Davies–Hatzel–Hickingbotham, *Quasi-isometries between graphs with variable edge lengths*** (10 Mar 2025) — fetched. Their Theorem 4: “For every $C \in \mathbb{N}$, there exist graphs G and H and a $(2,1)$ -quasi-isometry $\phi: V(G) \rightarrow V(H)$ such that, for every edge weighting $w: E(H) \rightarrow \mathbb{R}^+$, the map ϕ is not a $(1,C)$ -quasi-isometry from G to (H,w) .” Their Conjecture 8 is word-for-word NSS’s Conjecture 1.2 and is left **open** there.
- The writeup under review cites **no external results at all** — it is self-contained (the only imported idea is “the red/blue turn-cost mechanism from the supplied attempt”). So there is no citation to misquote. That is a scholarly weakness rather than a mathematical one: see below.

B.4 Computational check

Scripts in `verification_astra/scripts/2509.09031__00/`. Python 3 + networkx 3.6.1 + scipy 1.11.3.

1. `check_class_closure.py` — Lemma 1.1 on **all** connected graphs on ≤ 7 vertices from the networkx atlas: 995 connected graphs tested, **381 members of $\mathcal{C}$** , and for every member and every edge, contraction and subdivision both stay in $\mathcal{C}$. **0 failures**.
2. `check_closure_random.py` — same test on **20 000 randomly generated larger members of $\mathcal{C}$** (8–16 vertices: random subdivisions of K_{3,K_6} with pendant paths and attached cycles, plus random subdivided $G(n,p)$ graphs). **0 contraction failures, 0 subdivision failures**.
3. `verify_construction.py` — builds H as the incidence graph of $PG(2,3)$ (26 vertices, 4-regular, girth 6) as a computable stand-in for the Cayley graph H_g , 2-factorises it (Petersen, via Eulerian orientation + bipartite matching) into 26 red + 26 blue edges with all monochromatic cycles of length ≥ 6 , then builds G and H^* in full and checks p exhaustively over all pairs:

S	M	$ V(G) $	$ V(H^*) $	onto	$\max(d_{\{H^*\}} - d_G)$	$\max(d_G - (2d_{\{H^*\}} + 1))$
3	5	1248	1222	True	0	0
4	9	2392	2366	True	0	0
6	11	3042	3016	True	0	0

So p is onto and both quasi-isometry inequalities hold with room to spare, in both the writeup’s and NSS’s conventions. (Both maxima being exactly 0 means the bounds are tight, not violated.)

4. `verify_mechanism.py` — the Section-6 distance identities on the same instance ($S = 6$):
 - red edges: $d_G(u_R, v_R) = 6 = S$ for all 26 — claim confirmed;
 - blue edges: $d_G(u_R, v_R) = 8 = S+2$ for all 26 — claim confirmed;
 - monochromatic red 2- and 3-paths: 12 and 18 = St — confirmed;
 - monochromatic blue 2- and 3-paths: 14 and $\{18, 20\} = St+2$ — confirmed;
 - alternating paths: at $t = 1$, $d_G - (St+t-1) \in \{0, 2\} \geq 0$; at $t = 2$, $= 1 \geq 0$, so the lower bound (6.11) $d_G \geq St+t-1$ holds wherever the writeup’s hypothesis $2t < g/2$ is met. At $t = 3$ and $t = 4$ the bound *fails* on this instance ($d_G = 18 < 20$ at $t = 3$, and at $t = 4$ the endpoints are only at H -distance 2) — precisely because girth 6 breaks the uniqueness of the H -geodesic. This is a confirmation that the writeup’s girth hypotheses are correctly placed and genuinely load-bearing, not a counterexample.
 - best additive approximation of $d_G(u_R, v_R)$ by any edge weighting of this H : local search over all 52 weights reaches **1.7465** (natural choices $\lambda = S$ and $\lambda_e = d_G$ both give 2.0). Small, as expected, because the writeup’s bound only bites for girth $\gg 100$.
5. `verify_lemmas.py` — Lemma 2.2 (16 300 hypothesis instances over 4 000 random graphs, **0 violations**); the nine inequalities of block (6.5) at $g = 10\,000$ and $g = 100\,000$, all OK, and the block first holds in full at $g = 80$; stationarity of the alternating non-backtracking walk (max deviation from uniform = $0.000e+00$ over 6 steps); the cyclic-window counting identity; and the Section-4 partial-permutation recipe on all 1 456 nonempty reduced words of length < 7 .
6. `lp_turncost.py` / `lp_finite.py` — the quantitative core of §6.3–§6.4 recast as a linear program in the edge lengths λ and the error E : minimise E subject to $\lambda(P) \leq Ss+2+E$ for every monochromatic s -path and $\lambda(P) \geq Ss+s-1-E$ for every alternating s -path. On the finite 2-factorised H the optimum is

s	2	3	4	5	6
LP min E	0	0	0.5	1.0	1.5

i.e. exactly $(s-3)/2$, which is precisely the writeup’s conclusion $s-3 \leq 2E$. The writeup’s algebra is therefore reproduced numerically and exactly. Running the same LP on an **acyclic** (4-regular tree) model of the same local structure returns 0 for every s up to 7 — so the averaging over the *monochromatic cycles* in step (6.9) is genuinely load-bearing and is not a decoration. (The $s = 7$ entry on the finite graph drops back to 0 only because that toy H has monochromatic 6-cycles, which carry no simple 7-paths; the writeup requires all monochromatic cycles of length $\geq g > s$.)

What could not be checked by computation. The actual counterexample is not small: $g \geq 10\,000$, and $n = |Q_g|$ is the order of a subgroup of a product of symmetric groups indexed by all $\sim 4 \cdot 3^g$ short reduced words — astronomically large. Nothing about the asymptotic regime is testable directly, and Lemma 3.1 (a statement about arbitrary continuous maps between metric graphs) admits no brute-force analogue. Lemma 3.1 was checked by hand only, line by line; it is the single place where a subtle error would still be possible.

B.5 Caveats

1. **Coarse surjectivity is essential** and the counterexample is void without it. The writeup flags this but does not say why it is essential. NSS’s own definition does require it, so the writeup is safe — but a reader using a “quasi-isometric embedding” convention would wrongly reject the proof.
2. **Missing attribution.** The turn-cost mechanism of §6.1–§6.4 is close in spirit to Davies–Hatzel–Hickingbotham’s Theorem 4 (arXiv:2503.07448), which is exactly the fixed-target version of this obstruction, and the earlier attempt reproduced in `prompt.md` rediscovered that published theorem without noticing. The new writeup inherits the mechanism from that attempt and likewise cites nothing. The mechanisms are not identical (DHH orient a high-chromatic high-girth graph and use Burr’s theorem; here one 2-colours a 4-regular high-girth graph and averages over monochromatic cycles), but in a submitted paper §6 would have to cite DHH. This is a presentational defect, not a correctness one; the genuinely new content — Lemma 3.1 and the class $\mathcal{C}$ — is not in the literature I could find.
3. **Unstated hypothesis checks.** $n \geq 4$ in Lemma 3.1 is never verified in §6 (it holds); $\mathcal{C}$ is defined without the word “connected” although the conjecture asks for a class of connected graphs (contraction and subdivision preserve connectivity, so restricting $\mathcal{C}$ to connected graphs is harmless and should be stated); $D \geq 1$ is implicitly used in Step 1 ($10D+1 \leq 12D$).
4. **Scope.** The writeup itself notes that the variant of Conjecture 1.2 with the extra requirement of closure under edge deletion is untouched, as is the planar special case that DHH call out as the case of interest. $\mathcal{C}$ is a fairly artificial class (it is not closed under deletion, disjoint union, or minors, and it does not even contain all trees — a path of three consecutive branch vertices is not in $\mathcal{C}$). So the conjecture is refuted as stated, but the interesting instances are untouched. A referee of the source paper would likely accept the refutation and immediately re-ask the question for deletion-closed or minor-closed classes.
5. **Edge cases.** The argument never needs $A \geq 1$ or g small; the hypothesis $g \geq 10\,000$ is used only through block (6.5), which I verified first holds at $g = 80$, so the constants are far from tight but are sound.

B.6 Priority check

- Conjecture 1.2 of NSS (Sept/Oct 2025) is **verbatim** Conjecture 8 of Davies–Hatzel–Hickingbotham (arXiv:2503.07448, March 2025), where it is stated as the open weakening surviving their refutation of the fixed-target Conjecture 3. Both sources describe it as open.
- I searched the 2026 literature (arXiv listings and citing work: 2605.10068 “Coarse Menger property of quasi-minor excluded graphs”, 2608.10164 “Quasi-isometries, contractions, and intersection graphs”, 2603.21189, 2510.19602, 2510.14644, the NSS “Asymptotic structure” series I–V) and found **no paper resolving Conjecture 1.2 / Conjecture 8**. 2608.10164 (August 2026) works with subdivision/contraction operations and quasi-planarity but explicitly leaves open “whether M can always be chosen to be 1 by changing the target planar graph”.
- Neither the class $\mathcal{C}$ (“every two branch vertices joined by a thread”) nor a long-corridor rigidity lemma of the Lemma 3.1 kind turned up as a named result anywhere.
- **Self-contamination check:** none of the search hits traced back to this repository, the campaign catalog page, or the associated pull request. The two genuinely relevant sources (arXiv:2509.09031 and arXiv:2503.07448) are independent papers with named authors, dates and arXiv identifiers, and I opened both. So there is no ALREADY_KNOWN finding, and the novelty claim is not contaminated by our own artifacts.

B.7 Referee summary

I attacked this writeup expecting it to fail in one of the two ways the retry brief predicted — either the class $\mathcal{C}$ is not actually closed under the operations the conjecture requires, or the prose under-determines the object — and neither happened. $\mathcal{C}$ is well-defined and its closure under contraction and subdivision (Lemma 1.1) survived exhaustive testing on every connected graph with at most 7 vertices and 20 000 random larger members, with zero failures; the object is fully determined (H_g by the Cayley construction of §4, $S = g$, $M = 100g^2$, G_g by vertex-splitting, H_{g^+} as a subdivision of K_n), and the positive half — $p: G_g \rightarrow H_{g^+} \in \mathcal{C}$ is an onto (2,1)-quasi-isometry — is exactly right and was verified exhaustively on three explicit instances. Every distance identity Section 6 relies on reproduced numerically, the alternating-walk stationarity is exact, the cyclic-window counting is exact, all nine inequalities of block (6.5) hold with large margins, and when I recast the quantitative core as a linear program the optimum came out at exactly $(s-3)/2$, matching the writeup’s final inequality $s-3 \leq 2E$ to the digit; the same LP on an acyclic model returns 0, which confirms the monochromatic-cycle averaging is doing real work rather than decorating. The genuinely new step, the long-corridor rigidity Lemma 3.1 that converts an arbitrary additive quasi-isometry to an arbitrary member of $\mathcal{C}$ into edge lengths on a prescribed K_n skeleton, is where the whole argument lives and is the one part no computation can touch; I checked it clause by clause and every constant (12D, 14D, 20D, 34D, 50D, 63D, 25D, 30D) is correct under $S \geq 1000D$, but it is intricate enough that I record medium rather than high confidence. The defects I did find are all repairable in a sentence each — the case analysis behind “choosing a simple path in the image” in Lemma 1.1, the observation that the bad cycle in Lemma 2.2 is a whole component, the unstated derivations of the three distortion constants in Lemma 2.1, and the unchecked $n \geq 4$ hypothesis — so MINOR_GAPS, not MAJOR_GAP. Two things a journal would insist on: the writeup must cite Davies–Hatzel–Hickingbotham (arXiv:2503.07448) for the fixed-target version of the turn-cost obstruction, which it inherited from the previous attempt (that attempt had rediscovered their Theorem 4 without noticing), and it must state prominently that the coarsely-onto definition of quasi-isometry is not a convenience but a necessity — attach a length- 10^9 thread between every pair of G_g ’s branch vertices that lacks one and G_g embeds isometrically into a member of $\mathcal{C}$, so the counterexample evaporates under the quasi-isometric-embedding convention. NSS’s own definition does require coarse surjectivity, so the writeup is on safe ground. As for why the authors missed this: the object is not small — $n = |Q_g|$ for $g \geq 10\,000$ is astronomically large — the class $\mathcal{C}$ is artificial enough that nobody would test the conjecture against it, and the rigidity lemma is the real work. The conjecture is refuted as literally and correctly stated; the versions people actually care about (deletion-closed classes, planar graphs) remain open, and the writeup says so.

References

- [1] T. Nguyen, A. Scott and P. Seymour, Asymptotic structure. II. Path-width and additive quasi-isometry, [arXiv:2509.09031](#) (2025).
- [2] J. Davies, M. Hatzel and R. Hickingbotham, Quasi-isometries between graphs with variable edge lengths, [arXiv:2503.07448](#) (2025).